Demoscene AI 2022 Demo - One-Page Summary

Generated from repository evidence on 2026-04-08.

What it is

A single-page, music-synced demoscene web app built with Vite + TypeScript, Canvas 2D, and Web Audio API.
It plays a curated timeline of visual effects, transitions, and text cues over a soundtrack.

Who it is for

Primary persona: a demo creator/editor who wants to author or tweak audio-timed effect sections for a browser-based show.

What it does

- Runs a full-screen intro + section timeline with per-section effects, transitions, layering, and text cues.
- Synchronizes visuals to audio features (RMS/bass/mid/treble/beat) and supports manual beat injection via canvas taps.
- Supports many built-in effects plus WebGL2 effects with automatic Canvas2D fallback when WebGL2 is unavailable.
- Provides debug/editor controls, quality presets, framing modes, and URL-based rendering/performance tuning.
- Tracks live view counts using /api/views with GET/POST handling and KV-backed persistence when configured.
- Includes community workflows for submitting doodles and AI-generated effect ideas with moderation/review endpoints.

How it works (repo-evidenced architecture)

- Frontend app: src/main.ts wires canvas/UI, loads timeline config, initializes AudioPlayer, Timeline, and Renderer.
- Config layer: src/config/loadConfig.ts validates/normalizes raw JSON (audio, intro, sections, layers, transitions, cues).
- Playback core: src/timeline/timeline.ts computes intro/section mode, active transition windows, and active text cues by time.
- Audio analysis: src/audio/audioPlayer.ts uses Web Audio analyser data to emit beat + band energy features each frame.
- Rendering: src/renderer/renderer.ts and effect registry modules render the selected effect stack and transition blending.
- Service endpoints: api/views.ts, api/doodles.ts, api/effects.ts expose JSON APIs; they use createKvClients() for persistence.
- Data flow: timeline.release.json + song.mp3 -> load/normalize -> frame loop reads audio+time -> timeline state -> renderer ou

How to run (minimal)

1) Install Node.js 18+.
2) Run: npm install
3) Run: npm run dev
4) Open the local Vite URL and press Start.
5) Optional: replace public/song.mp3 and edit public/timeline.release.json for custom timing/effects.